NEET UG 2027: Master Biology Chapter Notes

Cell Biology, Cell Cycle, Animal Diversity & Anatomy (Frog & Cockroach)

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1 Cell Structure, Function & Biomolecules

Prokaryotic vs Eukaryotic Cell Architecture

1. Cell Theory Milestones:

- **M.J. Schleiden (1838)** observed plants; **Theodore Schwann (1839)** studied animal cells and noted presence of cell wall unique to plant cells.
- **Rudolf Virchow (1855)** modified cell theory: ***Omnis cellula-e cellula*** (all cells arise from pre-existing cells).

2. Prokaryotic Diagnostic Architecture:

- **Cell Envelope (3 Layers):** Outermost **Glycocalyx** (loose slime layer or rigid **capsule**) → Middle **Peptidoglycan cell wall** → Innermost semi-permeable **Plasma membrane**.
- **Mesosomes:** Invaginations of plasma membrane in the form of vesicles, tubules, and lamellae; increase surface area for enzymatic respiration, secretion, and replication.
- **Plasmids:** Small, circular, self-replicating extrachromosomal dsDNA; confer specific phenotypes like antibiotic resistance.
- **Inclusion Bodies:** Non-membrane bound storage granules free in cytoplasm (e.g., phosphate granules, cyanophycean granules, glycogen granules, gas vacuoles in blue-green and purple photosynthetic bacteria).

3. Eukaryotic Organelles & The Endomembrane System: The endomembrane system includes functionally coordinated organelles: **Endoplasmic Reticulum (ER), Golgi Apparatus, Lysosomes, and Vacuoles** (Mitochondria, Chloroplasts, and Peroxisomes are NOT part of it).

- **Fluid Mosaic Model (Singer & Nicolson, 1972):** Quasi-fluid lipid bilayer (phospholipids with hydrophilic heads outside, hydrophobic fatty acid tails inside) with integrated peripheral and integral proteins.
- **Rough ER (RER):** Studded with $80S$ ribosomes; active in **protein synthesis and secretion**.
- **Smooth ER (SER):** Major site for **lipid and steroid hormone synthesis**; detoxifies drugs in liver.
- **Golgi Complex (Camillo Golgi, 1898):** Flattened cisternae ($0.5 - 1.0\,\mu$m); convex forming **cis-face** receives vesicles from ER; concave maturing **trans-face** releases modified vesicles; crucial site for **glycosylation of proteins (glycoproteins)** and **lipids (glycolipids)**.
- **Lysosomes:** Formed by Golgi packaging; contain hydrolytic enzymes (lipases, proteases, carbohydrases) active at **acidic pH** (≈ 5.0).
- **Ribosomes (George Palade, 1953):** Non-membrane bound ribonucleoprotein granules ($70S = 50S + 30S$ in prokaryotes/organelles; $80S = 60S + 40S$ in eukaryotic cytoplasm).
- **Cilia & Flagella:** Possess $9 + 2$ axoneme microtubules; basal body derived from centrioles ($9 + 0$ cartwheel).

Enzyme Catalysis, Kinetics & Michaelis-Menten Dynamics

Enzymes are biocatalysts (mostly proteins; ribozymes are RNA enzymes like $23S$ rRNA):

- Lower the **activation energy** barrier without altering standard free energy change (ΔG) or equilibrium constant (K_{eq}).
- **Michaelis Constant (K_m):** Substrate concentration at which the reaction velocity reaches **half-maximal velocity** ($V_{\max}/2$).

$$K_m \propto \frac{1}{\text{Substrate Affinity of Enzyme}} \quad (\text{Lower } K_m \implies \text{Higher catalytic affinity}) \tag{1}$$

- **Competitive Inhibition:** Inhibitor closely resembles substrate in molecular structure and competes for active site:

$$\textbf{Competitive Inhibitor Effect:} \quad \mathbf{K_m} \text{ increases} \qquad \mathbf{V}_{\max} \text{ remains unchanged} \tag{2}$$

  Canonical NCERT Example: Inhibition of **succinate dehydrogenase by malonate** (resembles succinate).
- **Non-Competitive Inhibition:** Inhibitor binds to allosteric site: $\mathbf{V}_{\max}$ decreases, while $\mathbf{K_m}$ remains unchanged.
- **Enzyme Classes (IUBMB - 6 Classes):** (1) **Oxidoreductases**, (2) **Transferases**, (3) **Hydrolases**, (4) **Lyases** (cleave bonds without water), (5) **Isomerases**, (6) **Ligases** (join bonds with ATP hydrolysis).
- **Cofactors (Non-Protein Moieties):**
  - **Prosthetic Groups:** Tightly and permanently bound organic compounds (e.g., **Haem** in peroxidase and catalase).
  - **Coenzymes:** Transiently bound organic compounds derived from vitamins (e.g., **NAD and NADP** contain niacin).
  - **Metal Ions:** Form coordination bonds (e.g., Zn^{2+} in carboxypeptidase and carbonic anhydrase).

2 Cell Cycle & Cell Division: Mitosis & Meiosis

Cell Cycle Progression & Chromosome/DNA Mathematics

Standard eukaryotic cell cycle (human cell divides once in ≈ 24 hours; yeast divides in 90 minutes):

Interphase ($> 95\%$ of cycle): $\mathbf{G_1}$ (growth) $\longrightarrow$ **S** (DNA replication) $\longrightarrow$ $\mathbf{G_2}$ (spindle proteins)

M-Phase ($< 5\%$ of cycle): Prophase $\longrightarrow$ Metaphase $\longrightarrow$ Anaphase $\longrightarrow$ Telophase $\longrightarrow$ Cytokinesis

Quantitative Dynamics of Chromosome Number ($2n$) and DNA Content (C):

Stage	G_1 Phase	S Phase	G_2 Phase	Metaphase	Anaphase	Daughter Cells
Chromosome Count	$2n$	$2n$	$2n$	$2n$	**4n** (split centromeres)	$2n$
DNA Content	$2C$	**4C** (doubles)	$4C$	$4C$	$4C$	$2C$

Meiotic Substages: Prophase I Molecular Events

Meiosis reduces chromosome number by half ($2n \rightarrow n$). Prophase I is prolonged and divided into 5 distinct substages:

1. **Leptotene:** Chromosomes gradually become visible under light microscope; compaction of chromosomes continues.
2. **Zygotene:** Homologous chromosomes pair together (**Synapsis**), forming a nucleoprotein complex called **Synaptonemal Complex**. The paired complex is termed a **Bivalent** or **Tetrad**.
3. **Pachytene:** Non-sister chromatids of homologous chromosomes exchange genetic material (**Crossing Over**) at recombination nodules, catalyzed by the enzyme **Recombinase**.
4. **Diplotene:** Dissolution of synaptonemal complex; homologous chromosomes separate except at crossover sites, forming X-shaped configurations called **Chiasmata**. (Can remain arrested for months/years in oocytes of vertebrates, e.g., Dictyotene).
5. **Diakinesis: Terminalization of chiasmata**; chromosomes are fully condensed; nucleolus disappears and nuclear envelope breaks down.

Crucial Meiotic Checkpoints:

- **Anaphase I:** Homologous chromosomes separate to opposite poles; **sister chromatids remain attached at their centromeres** $\implies$ Reduction in chromosome number ($2n \rightarrow n$).
- **Anaphase II: Centromeres split**, allowing sister chromatids to move to opposite poles $\implies$ Resembles normal mitotic anaphase.

3 Animal Kingdom: Non-Chordates & Chordates Blueprint

Diagnostic Hallmarks of Major Animal Phyla

Phylum	Level of Org. / Symmetry	Coelom / Germ Layers	Diagnostic Features & Canonical Examples
Porifera (Sponges)	Cellular; Asymmetrical	Acoelomate; Diploblastic	**Choanocytes / Collar cells** line spongocoel; Water canal system; Spicules/spongin skeleton. *Sycon, Spongilla* (freshwater), *Euspongia* (bath sponge).
Cnidaria (Coelenterata)	Tissue; Radial symmetry	Acoelomate; Diploblastic	**Cnidoblasts / Nematocysts** on tentacles for defense/anchorage; Polyp and Medusa; **Metagenesis** in *Obelia. Hydra, Physalia* (Portuguese man-of-war), *Adamsia* (sea anemone).
Ctenophora (Comb jellies)	Tissue; Radial symmetry	Acoelomate; Diploblastic	Eight external rows of ciliated **comb plates** for locomotion; **Bioluminescence**; Strictly marine. *Pleurobrachia, Ctenoplana.*
Platyhelminthes (Flatworms)	Organ / Organ-system; Bilateral	**Acoelomate**; Triploblastic	Dorso-ventrally flattened body; Hooks and suckers in parasitic forms; **Flame cells (Protonephridia)** for osmoregulation. *Taenia* (tapeworm), *Fasciola* (liver fluke).
Aschelminthes (Roundworms)	Organ-system; Bilateral	**Pseudocoelomate**; Triploblastic	Circular cross-section; Muscular pharynx; Sexual dimorphism (female longer). *Ascaris, Wuchereria* (filarial worm), *Ancylostoma* (hookworm).
Annelida (Segmented worms)	Organ-system; Bilateral	**Coelomate**; Triploblastic	**Metameric segmentation**; Longitudinal/circular muscles; **Nephridia** for excretion; Closed circulation. *Nereis* (parapodia), *Pheretima* (earthworm), *Hirudinaria* (leech).
Arthropoda (Jointed legs)	Organ-system; Bilateral	Coelomate; Triploblastic	**Largest phylum** ($> 2/3$ species); Chitinous exoskeleton; **Malpighian tubules** for excretion; Statocysts; Open circulation. *Apis, Bombyx, Laccifer, Anopheles, Locusta, Limulus* (living fossil).
Mollusca (Soft-bodied)	Organ-system; Bilateral	Coelomate; Triploblastic	**Second largest phylum**; Calcareous shell; Unsegmented with head, muscular foot, and visceral hump; Mantle cavity with ctenidia; Rasping organ **Radula**. *Pila, Pinctada, Sepia, Octopus.*
Echinodermata (Spiny-skinned)	Organ-system; Radial (adult), Bilateral (larva)	Coelomate; Triploblastic	Endoskeleton of calcareous ossicles; **Water vascular system** (ambulacral system) for locomotion, food capture, and respiration; Excretory system absent. *Asterias* (starfish), *Echinus, Antedon.*
Hemichordata	Organ-system; Bilateral	Coelomate; Triploblastic	Worm-like marine; Proboscis, collar, and trunk; **Stomochord** in collar; Open circulation; Proboscis gland for excretion. *Balanoglossus, Saccoglossus.*
Chordata	Organ-system; Bilateral	Coelomate; Triploblastic	Fundamental hallmarks: (1) **Notochord**, (2) **Dorsal hollow nerve cord**, (3) **Paired pharyngeal gill slits**, (4) **Post-anal tail**.

4 Structural Organisation in Animals: Frog & Cockroach Anatomy

Cockroach Anatomy (*Periplaneta americana*) Prescribed Scope

- **Morphology:** Body divided into Head, Thorax, and Abdomen; covered by brown chitinous exoskeleton with hardened plates called **Sclerites** (dorsal **Tergites**, ventral **Sternites**, joined by flexible **arthrodial membrane**).
- **Head:** Triangular, hypognathous; formed by fusion of **6 embryonic segments**; bears a pair of compound eyes (each composed of $\approx$ **2000** hexagonal ommatidia providing mosaic vision with high sensitivity but low resolution).
- **Mouthparts (Biting and Chewing Type):** Labrum (upper lip), paired Mandibles (with grinding and incising teeth), paired Maxillae, Labium (lower lip), and median flexible lobe **Hypopharynx** (tongue).
- **Thorax:** Prothorax, Mesothorax, Metathorax. Forewings (**Tegmina** from mesothorax) are opaque, dark, leathery, and protective at rest; Hindwings (from metathorax) are transparent, membranous, and used in **flight**.
- **Abdomen:** Consists of **10 segments** in both sexes.
  - **Sexual Dimorphism:** Males bear a pair of unjointed **Anal Styles** on the 9th sternite (absent in females). Both sexes bear a pair of jointed **Anal Cerci** on the 10th segment.
- **Digestive System:** Alimentary canal: Mouth $\rightarrow$ Pharynx $\rightarrow$ Oesophagus $\rightarrow$ **Crop** (storage) $\rightarrow$ **Gizzard / Proventriculus** (has outer circular muscles and 6 thick chitinous teeth for grinding food).
  - **Hepatic / Gastric Caeca:** $6-8$ blind tubules at the junction of foregut and midgut; secrete digestive juice.
  - **Malpighian Tubules:** $100-150$ thin, yellow, filamentous tubules at the junction of midgut and hindgut; extract uric acid from hemolymph (**Uricotelic**).

- **Circulatory System:** Open type; heart consists of **13 funnel-shaped chambers** with ostia on either side.
- **Nervous System:** Consists of a series of fused ganglia: **3 in the thorax and 6 in the abdomen**. If head is cut off, cockroach survives for up to a week because the head holds only a bit of the nervous system!

Frog Anatomy (*Rana tigrina*) Prescribed Scope

- **Morphology:** Cold-blooded (**Poikilotherm**); amphibious; smooth, moist, slimy skin with mucus glands; camouflages (**Mimicry**). Body divided into **Head and Trunk** (neck and tail absent). Eyes are protected by a **Nictitating membrane**. Tympanum receives sound vibrations.
  - **Sexual Dimorphism:** Male frogs have sound-producing **Vocal Sacs** and a **Copulatory pad** on the first digit of the forelimb (both absent in female).
- **Respiration:** Cutaneous (skin in water and hibernation/aestivation), Buccopharyngeal cavity, and Pulmonary (lungs on land).
- **Circulatory System:** Closed, 3-chambered heart (**2 Atria and 1 Ventricle**).
  - Triangular **Sinus Venosus** joins the right atrium; receives blood from vena cava.
  - Ventricle opens ventrally into **Conus Arteriosus**.
  - **RBCs:** Nucleated, oval, and contain hemoglobin (unlike non-nucleated human RBCs).
  - Well-developed **Hepatic Portal** and **Renal Portal** systems.
- **Excretory System:** Paired compact, dark red **Mesonephric kidneys**; **Ureotelic** (excretes urea). Functional unit is uriniferous tubule. In males, ureters act as **Urogenital ducts** opening into the **Cloaca** (common chamber for digestive, excretory, and reproductive products).
- **Male Reproductive System:** Paired yellowish ovoid testes adhered to upper kidneys by double fold of peritoneum called **Mesorchium**. $10-12$ **Vasa efferentia** enter kidneys to run into **Bidder's Canal**, opening into urogenital duct.
- **Development:** External fertilization in water; indirect development via swimming aquatic larva called **Tadpole** (undergoes metamorphosis governed by thyroxine).